Monte Carlo Proxy Pricing

A textbook-style methodology for European, American, Asian, barrier, cliquet, basket Asian, SLV cliquet, basket cliquet, autocallable, and random-payoff proxies

Proxy Pricing Research Project

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Contents

Reader’s Guide

This document is written as a second-year undergraduate text. It assumes calculus, linear algebra, basic probability, and a willingness to read a few stochastic-process formulas. The goal is not to hide the implementation behind jargon. The goal is to show exactly how a Monte Carlo price label becomes a fast proxy.

The main chapters explain the workflow. The appendices give line-by-line proofs for the formulas used in the code. Every nontrivial implementation trick is classified as one of:

- an exact pricing identity,
- an unbiased Monte Carlo variance-reduction identity,
- a numerical approximation whose error must be validated.

Chapter 1

The Pricing Problem

1.1 Risk-neutral value

Let H be the payoff of a derivative at maturity or final settlement time T. Under the risk-neutral measure $\mathbb{Q}$, the time-t value is

$$V_t = B_t \, \mathbb{E}_{\mathbb{Q}}\left[\frac{H}{B_T} \,\middle|\, \mathcal{F}_t \right],$$

where $B_t = \exp(\int_0^t r_s \, \mathrm{d}s)$ is the money-market account. If the short rate is constant,

$$V_t = e^{-r(T-t)} \mathbb{E}_{\mathbb{Q}}[H \mid \mathcal{F}_t].$$

In the code, the expensive object is the conditional expectation. The proxy is a fast approximation of the map

$$x \longmapsto V(t, x),$$

where x is the state available at the valuation date.

1.2 What makes a good proxy state

A state is good when it is sufficient, low-dimensional, and scaled sensibly.

Definition 1.1 (Sufficient state)**.** A vector X_t is sufficient for pricing if, after conditioning on X_t, the distribution of all future cash flows no longer depends on earlier path history.

Examples:

- European option: $X_t = S_t$.
- Arithmetic Asian option: $X_t = (S_t, A_t)$, where A_t is the running sum of past fixings.
- American put: $X_t = S_t$, but the time index matters because of remaining exercise dates.
- Barrier option: $X_t = (S_t, I_t)$, where I_t says whether the barrier has already been hit.
- Single-name SLV cliquet: $X_t = (a_t, S_t, v_t)$, accrued coupon, spot, and variance.
- Basket SLV cliquet: $X_t = (a_t, S_t^{(1:3)}, v_t^{(1:3)})$, plus contract variant.

1.3 Error decomposition

The observed proxy error is not one number with one cause. A useful decomposition is

$$\widehat{V}_{\text{proxy}} - V_{\text{true}} = \underbrace{\widehat{V}_{\text{proxy}} - V_{\text{MC-label}}}_{\text{fitting error}} + \underbrace{V_{\text{MC-label}} - V_{\text{model}}}_{\text{training MC noise}} + \underbrace{V_{\text{model}} - V_{\text{true}}}_{\text{model/discretization error}} .$$

The experiments in this repository mostly study the first two terms under fixed model assumptions.

Chapter 2

Monte Carlo Label Construction

2.1 Plain estimator

For a state x, define the discounted path value

$$Y(x, Z) = e^{-r\tau} H(x, Z),$$

where Z represents all future random shocks. The target label is

$$V(x) = \mathbb{E}[Y(x, Z)].$$

With N simulated paths,

$$\widehat{V}_N(x) = \frac{1}{N} \sum_{n=1}^{N} Y(x, Z_n).$$

The estimator is unbiased when the paths have the correct law. Its standard error is estimated by

$$\widehat{\mathrm{SE}}(\widehat{V}_N) = \sqrt{\frac{1}{N(N-1)} \sum_{n=1}^{N} \left(Y_n - \overline{Y}\right)^2}.$$

2.2 Sobol and antithetics

Sobol points reduce integration error by filling the unit cube more evenly than independent random points. The implementation rounds the number of paths to a power of two because Sobol nets are naturally balanced in blocks of 2^m.

Antithetic pairing uses the fact that, for a standard normal vector Z,

$$Z \overset{d}{=} -Z.$$

The paired payoff

$$\frac{f(Z) + f(-Z)}{2}$$

has the same expectation as $f(Z)$. It often has lower variance for monotone or nearly monotone payoffs.

2.3 Likelihood-ratio importance sampling

State enrichment and importance sampling are different.

State enrichment changes the distribution of training states x_i. It does not change the path law for any individual conditional expectation.

Importance sampling changes the path density from p to q and multiplies by a likelihood ratio:

$$\mathbb{E}_p[h(Z)] = \mathbb{E}_q\left[h(Z)\frac{p(Z)}{q(Z)}\right].$$

For a normal mean shift, $q(z) = \phi(z - \theta)$, so

$$\frac{p(z)}{q(z)} = \exp\left(-\theta z + \frac{1}{2}\theta^2\right).$$

This identity is exact; its usefulness depends on variance.

2.4 Control variates

If C has known mean m_C, then

$$Y_\beta = Y - \beta(C - m_C)$$

has the same expectation as Y. The variance-minimizing coefficient is

$$\beta^\star = \frac{\mathrm{Cov}(Y, C)}{\mathrm{Var}(C)}.$$

Asian options use a geometric Asian control when the arithmetic payoff and geometric payoff are strongly correlated.

Chapter 3

Proxy Fitting Principles

3.1 Targets

Raw prices can be hard to fit because they range from nearly zero to almost linear. The code uses transformations such as:

$$\log(V + \varepsilon), \qquad \log \frac{V + \varepsilon}{B + \varepsilon}, \qquad \operatorname{logit} \frac{V - L}{U - L}.$$

Here B is a baseline, and L, U are known lower and upper bounds.

3.2 Exact regions

Never ask a regressor to learn something known analytically.

Examples:

- At maturity, use the payoff exactly.
- Deep Asian call in-the-money region: the positive part disappears and the value is linear.
- Cliquet global floor/cap tails: the final clipped payoff is already known.
- American/Bermudan exercise: impose $V = \max(\text{intrinsic}, \text{continuation})$.

3.3 Relative error floor

The reported relative error is

$$e_{\text{rel}} = \frac{|\widehat{V} - V|}{\max(|V|, \eta)}.$$

This is ordinary relative error when $|V| \geq \eta$, and it avoids meaningless exploding percentages when V is tiny.

Chapter 4

One-Dimensional Shape-Preserving Proxies

4.1 Why PCHIP is the default for 1D

For a single feature, PCHIP is a strong operational default:

- it interpolates labels exactly;
- it is local, so one noisy point does not create global oscillation;
- it preserves monotone shape when the data are monotone;
- it behaves well near exercise kinks, barrier transitions, and low-value tails.

4.2 Cubic Hermite form

On $[x_i, x_{i+1}]$, let $h_i = x_{i+1} - x_i$, $t = (x - x_i)/h_i$. A cubic Hermite segment is

$$\begin{aligned} H_i(t) = {} & y_i(2t^3 - 3t^2 + 1) + h_i d_i(t^3 - 2t^2 + t) \\ & + y_{i+1}(-2t^3 + 3t^2) + h_i d_{i+1}(t^3 - t^2). \end{aligned}$$

PCHIP chooses slopes d_i from neighboring secants in a way that prevents artificial extrema.

4.3 Akima, splines, and Bernstein

Akima interpolation is also local and robust to outliers. Natural cubic splines are smoother but can overshoot. Bernstein polynomials are shape-friendly on bounded intervals and were competitive for some barrier variants. For one-dimensional production defaults, the current rule is:

PCHIP first, compare Akima/Bernstein when the validation surface says so.

Chapter 5

Sparse Chebyshev and Higher Dimension

5.1 Chebyshev scaling

Chebyshev polynomials satisfy

$$T_n(\cos\theta) = \cos(n\theta),$$

so $|T_n(x)| \leq 1$ for $x \in [-1, 1]$. This is why features are scaled before fitting:

$$z_j = 2\frac{x_j - \ell_j}{u_j - \ell_j} - 1.$$

5.2 Ridge regression

Given a design matrix A, labels y, and penalty matrix Γ, ridge regression solves

$$\min_{\beta} \|A\beta - y\|_2^2 + \lambda\|\Gamma\beta\|_2^2.$$

The normal equation is

$$(A^\top A + \lambda\Gamma^\top\Gamma)\beta = A^\top y.$$

5.3 Why sparsity is necessary

A full tensor degree-p basis in D variables contains $(p+1)^D$ terms. With $D = 7$ and $p = 5$, that is 279,936 terms before fitting. Basket cliquet therefore needs sparse terms, payoff-aware features, or a slower safety method.

Chapter 6

Instrument Recipes

6.1 European option

State: S. Coordinate:

$$d_1 = \frac{\log(S/K) + (r - q + \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}}.$$

Training labels use shifted Sobol terminal GBM paths. Validation uses Black-Scholes:

$$C = Se^{-q\tau}\Phi(d_1) - Ke^{-r\tau}\Phi(d_2), \qquad d_2 = d_1 - \sigma\sqrt{\tau}.$$

Default proxy: log price fitted by PCHIP in d_1.

6.2 Arithmetic Asian option

State: S and running sum A. With m future fixings and total N,

$$K_{\text{adj}} = \frac{NK - A - S}{m}.$$

The call payoff can be rewritten as

$$\left(\frac{A + S + \sum_{j=1}^{m} SG_j}{N} - K\right)_+ = S\frac{m}{N}\left(\frac{1}{m}\sum_{j=1}^{m} G_j - \frac{K_{\text{adj}}}{S}\right)_+.$$

Thus the tested monthly call can be represented by an adjusted-moneyness coordinate. Training uses Sobol antithetics, likelihood shifts where useful, and a geometric Asian control variate.

6.3 Ten-asset basket Asian

State: ten spots plus the running basket sum. The proxy starts from a moment-matched lognormal baseline for the final arithmetic basket average, then fits a sparse Chebyshev/PCA correction and a PCHIP residual calibration. PCA summarizes cross-sectional composition without throwing away the level and running-sum information.

6.4 American and Bermudan options

At each exercise date k,

$$V_k(S) = \max\big(g_k(S),\, e^{-r\Delta t}\mathbb{E}[V_{k+1}(S_{k+1}) \mid S_k = S]\big)\,.$$

The continuation value is estimated by one-step Sobol transitions and fitted with PCHIP. The max with intrinsic value is imposed exactly.

6.5 Barrier option

For a zero-rebate barrier,

$$V_{\text{in}} + V_{\text{out}} = V_{\text{vanilla}}.$$

Discrete monitoring checks simulated observation dates. Continuous monitoring uses Brownian-bridge survival between observations. For a lower log barrier b, endpoints $x, y > b$, and variance $\sigma^2\Delta t$,

$$P(\text{survive} \mid x, y) = 1 - \exp\left(-\frac{2(x-b)(y-b)}{\sigma^2\Delta t}\right).$$

6.6 Cliquet and SLV cliquet

Cliquet payoffs are bounded by local and global caps/floors. If accrued value a and m remaining coupons imply the final total is certainly below the global floor or above the global cap, the value is exact.

For single-name SLV cliquet, the state is (a, S, v). The proxy uses exact tails, bounded-logit targets, and local/sparse Chebyshev features. For three-asset basket cliquet, the fitted proxy is not universally enough for strict max-error control; hard order-statistic cases use a cached Sobol/LR safety proxy.

6.7 Autocallable and random payoff

Autocallable valuation is naturally date-indexed. The state is spot at an observation date; the cash-flow rule determines whether the note redeems early or continues. Random payoff tests are synthetic one-dimensional payoffs used to stress interpolation under different shapes.

Chapter 7

Validation and Timing

7.1 Independent benchmark principle

Training labels and benchmark labels use separate random streams. Benchmarks use more paths, closed forms, finite differences, or tree methods when available.

7.2 Current reporting

The timing report separates:

- training sample generation;
- proxy fitting after labels are available;
- option-family wall time;
- average wall time per scenario.

The reduced-budget smoke accuracy columns in the timing report are diagnostic only. The product-specific markdown summaries remain the full-budget accuracy source.

Appendix A

Line-by-Line Proofs: Monte Carlo

A.1 Unbiased sample mean

Line 1. Define $Y = e^{-r\tau} H$ and $V = \mathbb{E}_{\mathbb{Q}}[Y]$.

Line 2. Define $\widehat{V}_N = N^{-1} \sum_{n=1}^{N} Y_n$.

Line 3. Apply linearity:

$$\mathbb{E}[\widehat{V}_N] = \mathbb{E}\left[\frac{1}{N}\sum_{n=1}^{N} Y_n\right] = \frac{1}{N}\sum_{n=1}^{N} \mathbb{E}[Y_n] = \frac{1}{N}\sum_{n=1}^{N} V = V.$$

Line 4. If paths are independent with variance σ_Y^2,

$$\mathrm{Var}(\widehat{V}_N) = \mathrm{Var}\left(\frac{1}{N}\sum_{n=1}^{N} Y_n\right) = \frac{1}{N^2}\sum_{n=1}^{N} \mathrm{Var}(Y_n) = \frac{\sigma_Y^2}{N}.$$

Line 5. Replace σ_Y^2 by the sample variance to estimate standard error.

A.2 Likelihood-ratio shift

Line 1. Start with $V = \int h(z)p(z)\,\mathrm{d}z$.

Line 2. Choose a proposal q with $q(z) > 0$ wherever $h(z)p(z) \neq 0$.

Line 3. Multiply and divide by q:

$$V = \int h(z)\frac{p(z)}{q(z)}q(z)\,\mathrm{d}z = \mathbb{E}_q[h(Z)L(Z)], \qquad L(Z) = \frac{p(Z)}{q(Z)}.$$

Line 4. For $p(z) = \phi(z)$, $q(z) = \phi(z - \theta)$,

$$L(z) = \frac{\exp(-z^2/2)}{\exp(-(z-\theta)^2/2)} = \exp\left(-\theta z + \frac{\theta^2}{2}\right).$$

Line 5. For independent shocks, likelihood ratios multiply:

$$L(z_1, \dots, z_m) = \prod_{j=1}^{m} \exp\left(-\theta_j z_j + \frac{\theta_j^2}{2}\right).$$

A.3 Antithetic estimator

Line 1. Standard normal symmetry gives $Z \overset{d}{=} -Z$.

Line 2. Define $A = (f(Z) + f(-Z))/2$.

Line 3. Then

$$\mathbb{E}[A] = \frac{\mathbb{E}[f(Z)] + \mathbb{E}[f(-Z)]}{2} = \frac{\mathbb{E}[f(Z)] + \mathbb{E}[f(Z)]}{2} = \mathbb{E}[f(Z)].$$

Line 4. The variance is

$$\operatorname{Var}(A) = \frac{1}{2}\operatorname{Var}(f(Z)) + \frac{1}{2}\operatorname{Cov}(f(Z), f(-Z)).$$

Line 5. The pairing helps whenever the covariance term is negative.

Appendix B

Line-by-Line Proofs: GBM and European Options

B.1 GBM solution

Line 1. Start from $\mathrm{d}S_t/S_t = (r-q)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t$.

Line 2. Apply Ito's lemma to $f(S) = \log S$:

$$\mathrm{d}\log S_t = \frac{1}{S_t}\,\mathrm{d}S_t - \frac{1}{2}\frac{1}{S_t^2}(\,\mathrm{d}S_t)^2.$$

Line 3. Since $(\,\mathrm{d}S_t)^2 = \sigma^2 S_t^2\,\mathrm{d}t$,

$$\mathrm{d}\log S_t = (r - q - \tfrac{1}{2}\sigma^2)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t.$$

Line 4. Integrate over $[t, T]$:

$$\log(S_T/S_t) = (r - q - \tfrac{1}{2}\sigma^2)\tau + \sigma(W_T - W_t).$$

Line 5. Use $W_T - W_t = \sqrt{\tau}Z$, $Z \sim N(0,1)$, and exponentiate.

B.2 Black-Scholes call

Line 1. Risk-neutral pricing:

$$C = e^{-r\tau}\mathbb{E}[(S_T - K)_+].$$

Line 2. Split the positive part:

$$\mathbb{E}[(S_T - K)_+] = \mathbb{E}[S_T \mathbf{1}_{\{S_T > K\}}] - K\mathbb{Q}(S_T > K).$$

Line 3. The event $S_T > K$ is $Z > -d_2$, so $\mathbb{Q}(S_T > K) = \Phi(d_2)$.

Line 4. Completing the square gives

$$\mathbb{E}[S_T \mathbf{1}_{\{S_T > K\}}] = Se^{(r-q)\tau}\Phi(d_1).$$

Line 5. Substitute and discount:

$$C = Se^{-q\tau}\Phi(d_1) - Ke^{-r\tau}\Phi(d_2).$$

Appendix C

Line-by-Line Proofs: Asian State Reduction

C.1 Adjusted moneyness

Line 1. At a valuation date, write future fixings as $SG_1, \ldots, SG_m$.

Line 2. The arithmetic average is

$$\bar{S} = \frac{A + S + S\sum_{j=1}^{m} G_j}{N}.$$

Line 3. Define

$$K_{\text{adj}} = \frac{NK - A - S}{m}.$$

Line 4. Rewrite the call payoff:

$$(\bar{S} - K)_+ = \frac{1}{N}\left(A + S + S\sum_{j=1}^{m} G_j - NK\right)_+ = \frac{m}{N}\left(S\frac{1}{m}\sum_{j=1}^{m} G_j - K_{\text{adj}}\right)_+.$$

Line 5. Factor out $S > 0$:

$$(\bar{S} - K)_+ = S\frac{m}{N}\left(\frac{1}{m}\sum_{j=1}^{m} G_j - \frac{K_{\text{adj}}}{S}\right)_+.$$

Line 6. Under GBM the growth factors G_j do not depend on the current spot level S.

Line 7. Therefore $V(S, A) = SF(K_{\text{adj}}/S, t)$, which justifies a one-dimensional adjusted-moneyness proxy for the tested call setup.

Appendix D

Line-by-Line Proofs: Barriers

D.1 In/out parity

Line 1. Split all paths into two disjoint events: hit and no hit.

Line 2. On hit paths, the knock-in pays the vanilla payoff and the knock-out pays zero.

Line 3. On no-hit paths, the knock-out pays the vanilla payoff and the knock-in pays zero.

Line 4. Therefore, path by path,

$$H_{\text{in}} + H_{\text{out}} = H_{\text{vanilla}}.$$

Line 5. Discount and take expectations:

$$V_{\text{in}} + V_{\text{out}} = V_{\text{vanilla}}.$$

D.2 Brownian-bridge lower-barrier survival

Line 1. Condition on log endpoints $X_0 = x$, $X_{\Delta t} = y$, both above lower barrier b.

Line 2. By the reflection principle, the conditional probability of crossing below b is

$$\exp\left(-\frac{2(x-b)(y-b)}{\sigma^2 \Delta t}\right).$$

Line 3. Survival is one minus crossing:

$$P_{\text{surv}} = 1 - \exp\left(-\frac{2(x-b)(y-b)}{\sigma^2 \Delta t}\right).$$

Line 4. Conditional survival over many intervals is the product of interval survival probabilities.

Appendix E

Line-by-Line Proofs: Exercise and Cliquet Bounds

E.1 American recursion

Line 1. At final date M, $V_M(S) = g_M(S)$.

Line 2. Suppose V_{k+1} is known.

Line 3. Continuing from date k has value

$$C_k(S) = e^{-r\Delta t}\mathbb{E}[V_{k+1}(S_{k+1}) \mid S_k = S].$$

Line 4. Stopping immediately has value $g_k(S)$.

Line 5. The holder chooses the larger:

$$V_k(S) = \max(g_k(S), C_k(S)).$$

Line 6. Backward induction proves optimality at every date.

E.2 Cliquet exact tails

Line 1. Let current accrued return be a, with m future local coupons.

Line 2. Each future coupon lies in $[\ell, u]$.

Line 3. Therefore final accumulated return lies in

$$a + \sum_{j=1}^{m} c_j \in [a + m\ell,\ a + mu].$$

Line 4. If $a + mu$ is below the global floor, the payoff is exactly the global floor.

Line 5. If $a + m\ell$ is above the global cap, the payoff is exactly the global cap.

Line 6. These states need no regression and no simulation.

Appendix F

Line-by-Line Proofs: PCHIP

F.1 Unique Hermite cubic

Line 1. A cubic has four coefficients.

Line 2. Endpoint value constraints give two linear equations.

Line 3. Endpoint slope constraints give two more linear equations.

Line 4. The Hermite basis constructs one solution:

$$H(t) = y_i h_{00}(t) + h_i d_i h_{10}(t) + y_{i+1} h_{01}(t) + h_i d_{i+1} h_{11}(t).$$

Line 5. If two cubics satisfied all four conditions, their difference would have double roots at both endpoints.

Line 6. A nonzero cubic cannot have four roots counting multiplicity, so the solution is unique.

F.2 Why PCHIP avoids artificial extrema

Line 1. Let $\delta_i = (y_{i+1} - y_i)/h_i$.

Line 2. If neighboring secants change sign, PCHIP sets the node slope to zero.

Line 3. If neighboring secants have the same sign, PCHIP uses the weighted harmonic mean

$$d_i = \frac{w_1 + w_2}{w_1/\delta_{i-1} + w_2/\delta_i}.$$

Line 4. This slope has the same sign as the neighboring secants and lies between their magnitudes.

Line 5. On a monotone interval, the derivative of the Hermite cubic is a quadratic whose normalized endpoint slopes satisfy the Fritsch-Carlson monotonicity inequalities.

Line 6. Therefore the derivative does not change sign inside the interval.

Line 7. Hence the interpolant cannot create a new local maximum or minimum between two monotone data points.

Appendix G

Line-by-Line Proofs: Sparse Chebyshev Ridge

G.1 Chebyshev boundedness

Line 1. Every $x \in [-1, 1]$ can be written as $x = \cos\theta$.

Line 2. Define $T_n(x) = \cos(n\theta)$.

Line 3. Since $|\cos(n\theta)| \leq 1$,

$$|T_n(x)| \leq 1, \qquad x \in [-1, 1].$$

Line 4. Scaling state variables to $[-1, 1]$ keeps basis columns numerically bounded.

G.2 Ridge normal equation

Line 1. Objective:

$$J(\beta) = \|A\beta - y\|_2^2 + \lambda\|\Gamma\beta\|_2^2.$$

Line 2. Differentiate:

$$\nabla J(\beta) = 2A^\top(A\beta - y) + 2\lambda\Gamma^\top\Gamma\beta.$$

Line 3. Set the gradient to zero:

$$A^\top A\beta - A^\top y + \lambda\Gamma^\top\Gamma\beta = 0.$$

Line 4. Rearrange:

$$(A^\top A + \lambda\Gamma^\top\Gamma)\beta = A^\top y.$$

Line 5. The left matrix is positive semidefinite, and ridge makes unsupported directions stable.

Appendix H

Line-by-Line Proofs: Bounded Targets and Metrics

H.1 Logit bounded target

Line 1. Suppose the price must lie in $[L, U]$.

Line 2. Fit an unconstrained number y and map it through the logistic function:

$$u(y) = \frac{1}{1 + e^{-y}}.$$

Line 3. Because $e^{-y} > 0$, $0 < u(y) < 1$.

Line 4. Rescale:

$$\widehat{V} = L + (U - L)u(y).$$

Line 5. Therefore $L < \widehat{V} < U$.

H.2 Relative-error floor

Line 1. Define

$$e_{\text{rel}} = \frac{|\widehat{V} - V|}{\max(|V|, \eta)}, \qquad \eta > 0.$$

Line 2. If $|V| \geq \eta$, this is ordinary relative error.

Line 3. If $|V| < \eta$, then

$$e_{\text{rel}} = \frac{|\widehat{V} - V|}{\eta}.$$

Line 4. Thus a nearly zero benchmark cannot create an infinite percentage error.

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